

AfyaPlus Executive Evidence Brief

Decision: Approve controlled, quality-gated use of openai/gpt-4o-mini for the evaluated AfyaPlus workflows. Do not treat simulation results or this static brief as proof of production readiness or regulatory compliance.

Executive Summary

AfyaPlus's generative layer passed clinical evaluation on openai/gpt-4o-mini (15/15 questions, 7/7 quality gates, 100% grounding compliance) and is projected to run at roughly one-tenth of a US cent per 30-day workload day under the current mixed-model traffic assumption. The evaluated openai/gpt-4o alternative failed the required grounding-compliance gate and is not eligible for routing. Two systemic risks require engineering follow-up before wider deployment: simulated quality drift appears by Month 3, and there is currently no live production quality signal to confirm it does not also occur with real traffic.

Quality Performance Breakdown

- **Measured:** openai/gpt-4o-mini completed **15/15 questions**, passed **7/7 clinical gates**, scored **5.0000/5 overall**, and achieved **100% grounding compliance** [E1, E2]. In plain terms: every answer stayed faithful to the documented policy/clinical source, and no answer required a safety override.
- **Measured:** openai/gpt-4o completed **15/15 questions** but achieved only **93.33% grounding compliance** — one answer in fifteen was not fully supported by the retrieved reference — so it failed the required **100%** gate and is not eligible for cost-led routing [E1, E2].

Cost & Efficiency Analysis

- **Measured:** Per-request cloud cost is **USD 0.00008024** for gpt-4o-mini versus **USD 0.00126867** for gpt-4o [E7].
- **Simulated:** The 30-day mixed-model workload contains **9,000 requests** and projects **USD 3.39612000 / KES 439.80**, consuming **94.34%** of both the daily and monthly budget caps and producing a **WARNING** status on each [E4].
- **Simulated:** All-mini routing preserves every recorded clinical gate and projects **USD 0.72216000** over 30 days, saving **USD 2.67396000 (78.74%)** against the mixed baseline [E5].
- **Estimated (illustrative assumption, not measured):** A human-in-the-loop clinical reviewer, at an assumed **USD 30/hour** spending **2 minutes** validating each response, costs roughly **USD 1.00 per response** — four to five orders of magnitude more than either model's per-request cloud cost above. This is a planning illustration, not a costed staffing proposal: no human-in-the-loop review program currently exists in this system, and the assumed rate/throughput have not been validated against real AfyaPlus reviewer capacity.

Systemic Operational Risks

- **Simulated:** Month 2 first detected operational drift of **+563.582 ms latency** and **+103.408 tokens** per request. Month 3 added actionable quality drift: ROUGE-L **-0.110996**, overall score **-0.594667**, and grounding compliance **-20 percentage points** versus reference [E3].
- **Live-only:** Current API health is intentionally not frozen into this report. The authenticated dashboard probes the real /health endpoint live and reports **UNKNOWN** when the service cannot be reached [E6].
- **Live-only:** The dashboard's exception counter (4xx/5xx responses on the real Chat API) is read live at request time, not simulated — it is intentionally absent from this static brief for the same reason [E6].

Actionable Engineering Roadmap

- **Put safety before price** (clinical justification): retain the all-gates-pass requirement and route only to models that satisfy it, even though gpt-4o would otherwise be viable on cost alone.
- **Adopt openai/gpt-4o-mini for this evaluated scope** (financial justification): simulated all-mini routing preserves every recorded gate and saves **USD 2.67396000 (78.74%)** over 30 days against the mixed baseline [E5].
- **Treat any actionable quality-drift alert as a release hold** (safety justification): investigate, re-evaluate, and restore gate compliance before expanding traffic, rather than absorbing a Month-3-style quality regression silently.
- **Require human clinical and governance review before broader deployment** (compliance justification): these results do not establish legal compliance, patient safety in every scenario, or end-to-end production readiness.

Source register

- [E1] evaluation/model_comparison_summary.csv — columns model, total_questions, successful_questions, overall, all_gates_passed.
- [E2] evaluation/quality_gate_log.csv — columns model, gate, threshold, observed, passed.
- [E3] drift/drift_trend_table.csv — columns month, column, mean_change, drift_detected, requires_action.
- [E4] cost/cost_projection_30d.csv — overall-row columns projected_requests, projected_cost_usd, projected_cost_kes, daily_budget_usd, budget_utilization, budget_status, monthly_budget_usd, monthly_budget_utilization, monthly_budget_status.
- [E5] cost/structural_savings_analysis.csv — columns status, selected_model, quality_preserved, proposed_30d_usd, savings_usd, savings_percent.
- [E6] dashboard/health.py — probe_health, parse_exception_count; dashboard/README.md — "Architecture and failure boundaries".
- [E7] cost/cost_per_request_comparison.csv — columns model, average_cost_per_request_usd.